Introduction to Optimal Control Problem

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Laboratory 2

In this laboratory, rotational maneuvers of a rigid satellite will be studied.

Laboratory 2

First, the Rest-to-Rest rotational maneuver of a rigid satellite will be studied (Page 80 – Ben Asher book).

Rest-to-Rest rotational maneuver of a rigid satellite

Problem

$$\min J(x_1, x_2) = \int_0^1 x_2'^2(t)\, dt,$$

$$\text{s.t. } x_1'(t) - x_2(t) = 0,$$

and the initial and final conditions:

$$x_1(0) = 0,$$
$$x_2(0) = 0,$$
$$x_1(1) = 1,$$
$$x_2(1) = 0.$$

Rest-to-Rest rotational maneuver of a rigid satellite

Important

Physically, this problem represents the control of a satellite's rotational maneuver, with $x_1(t)$ and $x_2(t)$ being the angular position and angular velocity, respectively, controlled by an external torque.

Rest-to-Rest rotational maneuver of a rigid satellite

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onsidering the physical meaning of the problem, one can avoid CV formulations and obtain an Optimal Control problem. We make the following variable change:

$$x_2'(t) = u(t)$$

Rest-to-Rest rotational maneuver of a rigid satellite

The new problem formulation is:

$$\min J(x_1, x_2) = \int_0^1 u^2(t)\, dt,$$

$$\text{s.t.} \quad x_1'(t) - x_2(t) = 0,$$
$$x_2'(t) - u(t) = 0,$$

and the boundary conditions:

$$x_1(0) = 0,$$
$$x_2(0) = 0,$$
$$x_1(1) = 1,$$
$$x_2(1) = 0.$$

Rest-to-Rest rotational maneuver of a rigid satellite

Using the standard optimal control notation:
State variables: $x_1(t)$, $x_2(t)$
Control variable: $u(t)$

Laboratory

Question 1

Solve the previous problem.

Pyomo

Using Pyomo.

Laboratory

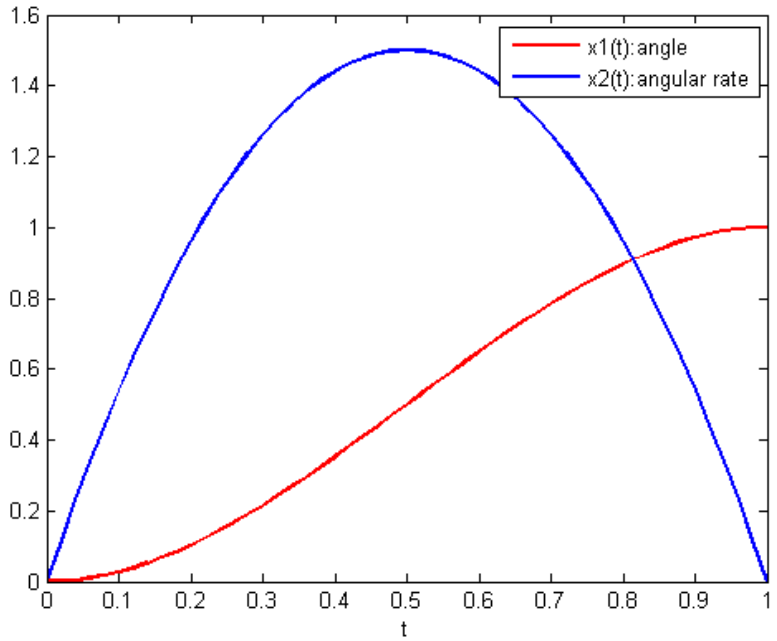
Solution

SOLUTION

Laboratory

Problem 1: Solution

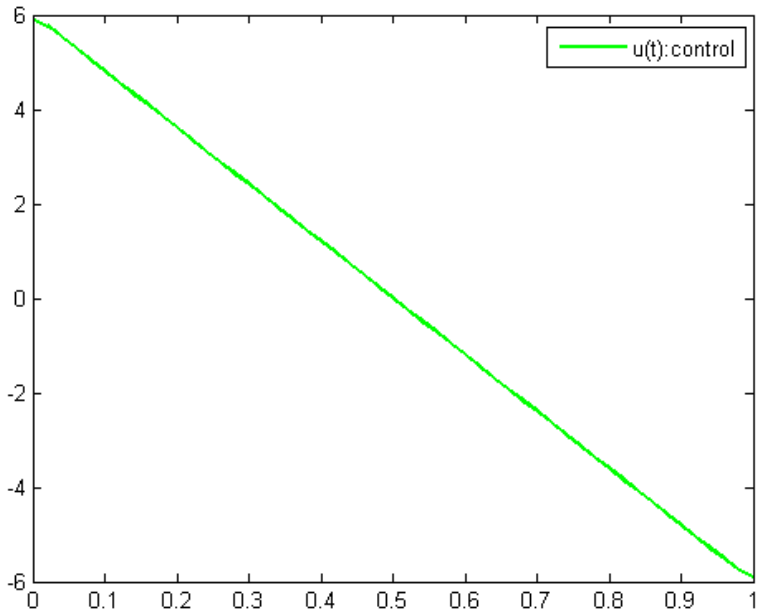
Plot of the obtained state variables:



Laboratory

Problem 1: Solution

And the control:



Laboratory 2

Next, the so-called Spin-Up maneuver of a rigid satellite will be studied (Page 82 – Ben Asher book).

Problem 2

Consider the problem:

$$\min J(x_1, x_2) = \int_0^1 x_2'^2(t)\, dt,$$

subject to the differential constraint:

$$x_1'(t) - x_2(t) = 0,$$

and the initial and final conditions:

$$\begin{aligned} x_1(0) &= 0, \\ x_2(0) &= 0, \\ x_2(1) &= 1. \end{aligned}$$

Laboratory

Question 1

Solve the previous problem.

Pyomo

Using Pyomo.

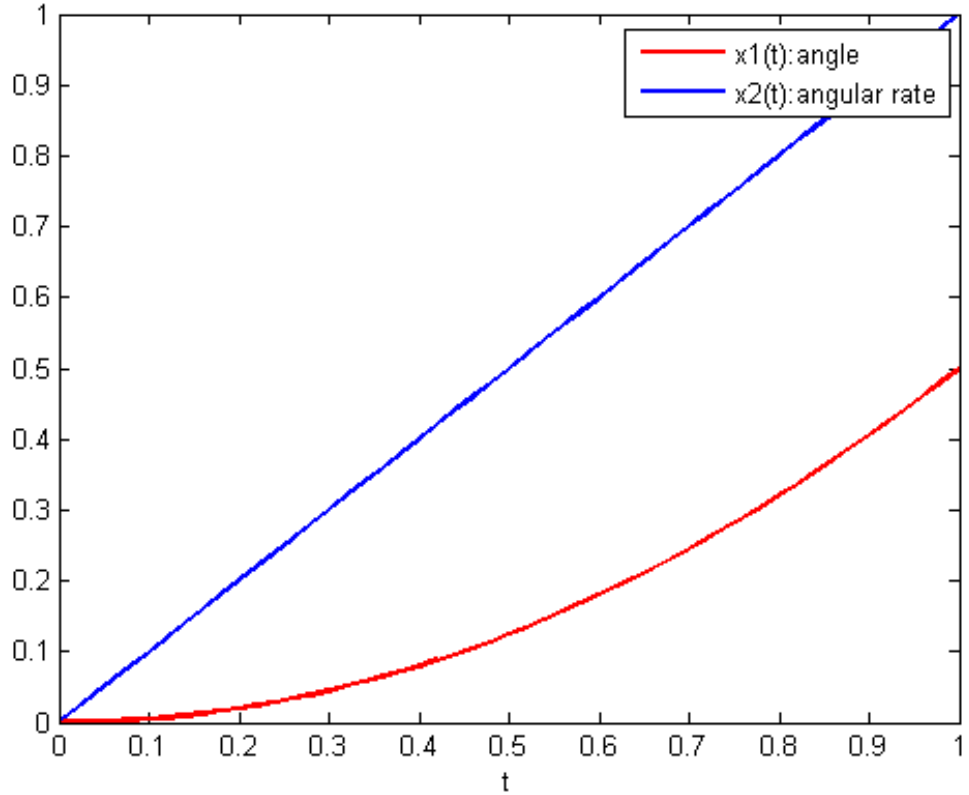
Laboratory

SOLUTION

Laboratory

Problem 2: Solution

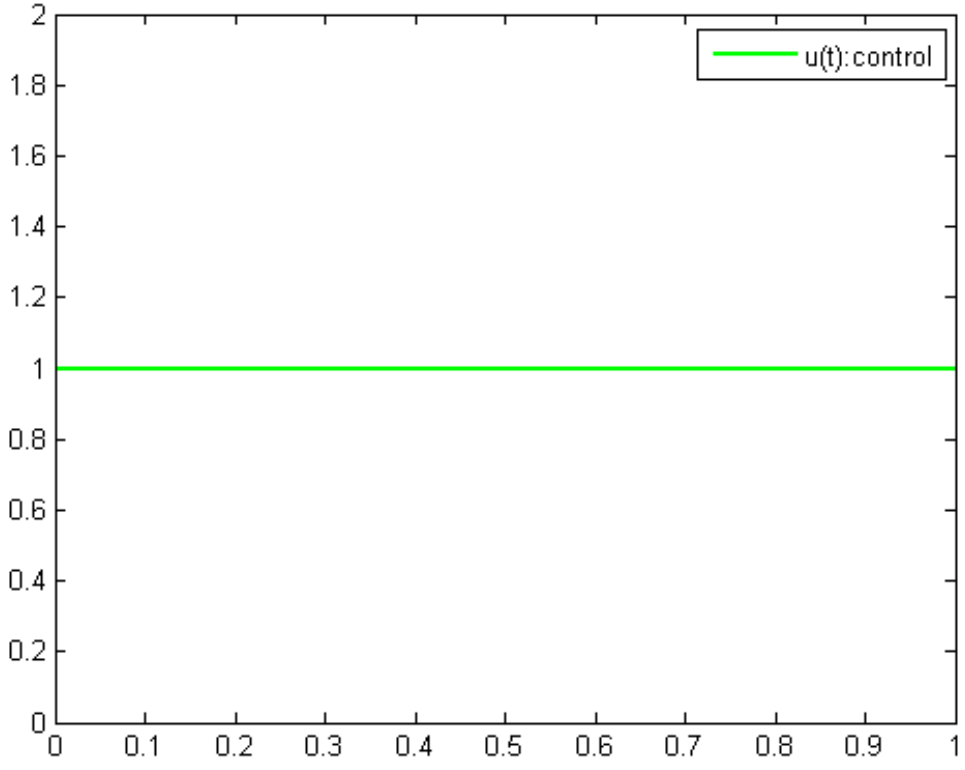
Plot of the obtained state variables:



Laboratory

Problem 2: Solution

And for the control:



Laboratory

Problem 1 and 2: Solution

Compare the solutions obtained.